Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6097491812

COMPUTER SCIENCE **2210/23**

Paper 2 Problem-solving and Programming **October/November 2018**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

No calculators allowed.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

DO NOT ATTEMPT TASKS 1, 2 AND 3 in the pre-release material; these are for information only.

You are advised to spend no more than **40 minutes** on **Section A** (Question 1).

No marks will be awarded for using brand names of software packages or hardware.

Any businesses described in this paper are entirely fictitious.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

The maximum number of marks is 50.

This document consists of **11** printed pages and **1** blank page.

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Section A

You are advised to spend no longer than 40 minutes answering this section.

Here is a copy of the pre-release material.

DO NOT attempt Tasks 1, 2 and 3 now.

Use the pre-release material and your experience from attempting the tasks before the examination to answer Question 1.

Pre-release material

A holiday park has a number of log cabins that it rents by the week as shown in the table:

Name	Capacity	Peak	Off-peak
Hetty	4	$400.00	$250.00
Poppy	4	$400.00	$250.00
Blue Skies	4	$500.00	$350.00
Bay View	6	$650.00	$500.00
Happy Days	6	$695.00	$550.00
Summer Joy	6	$800.00	$600.00
Walkers' Rest	8	$950.00	$750.00
Bertie	8	$1050.00	$850.00
Green Forest Lodge	10	$1200.00	$950.00
Coppice Lodge	10	$1500.00	$1150.00

The capacity represents the maximum number of occupants for each log cabin. A program is needed to record and store bookings. Log cabins can only be booked from weeks labelled on the calendar as weeks 23 to 39, inclusive. Peak rates operate for weeks 27 to 35, inclusive, and off-peak rates apply for weeks 23 to 26 and weeks 36 to 39, inclusive.

Write and test a program or programs.
- Your program or programs must include appropriate prompts for the entry of data.
- Error messages and other output need to be set out clearly and understandably.
- All arrays, variables, constants and other identifiers must have meaningful names.

You will need to complete these **three** tasks. Each task must be fully tested.

TASK 1 – Setting up the bookings system.
Write a program, using arrays, to identify each log cabin, its capacity, cost and whether or not it has been booked for each week. Identify each week by a number ranging from week 23 to 39.

TASK 2 – Taking a booking.
Extend the program to:
- Identify and display which weeks are available for each log cabin, and its capacity.
- Input the log cabin, number of weeks and start week for the booking.
- Generate a unique booking code for the week(s) and log cabin chosen.
- Store the unique booking code in your array
  (multiple week bookings will need the booking code stored multiple times).
- Calculate and output the cost of the booking.

TASK 3 – Applying a special offer.
Amend the program to apply a 10% discount to any booking of three weeks or more. Output the original cost and the discounted cost of the booking.

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1 **(a)** All arrays, variables, constants and other identifiers should have meaningful names.

(i) State the name, data type and use of **three** arrays you have created for **Task 1**.

Array 1 name .....

Data type .....

Use .....

.....

Array 2 name .....

Data type .....

Use .....

.....

Array 3 name .....

Data type .....

Use .....

.....

[6]

(ii) State the name, data type and use for **one** variable you used in **Task 2**.

Variable name .....

Data type .....

Use .....

.....

[2]

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[Turn over

(b) Write an algorithm to show how your program finds and displays the weeks available for each log cabin and its capacity (part of **Task 2**), using **either** pseudocode, programming statements or a flowchart.

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(c) Describe how you could validate the input to identify a log cabin in **Task 2**. State **one** valid and **one** invalid item of data to test your validation method.

Validation method .....

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Valid test data .....

Invalid test data .....

[4]

(d) Explain how your program performs **Task 3**. Any programming statements used in your answer must be fully explained.

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.....[4]

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Section B

2 Describe, giving an example for each, the following data types used in programming.

Integer

Description .....

.....

Example .....

String

Description .....

.....

Example .....

[4]

3 Give an example of a pseudocode statement or statements to perform each of the following functions.

A condition controlled loop .....

.....

.....

A conditional statement .....

.....

.....

Totalling .....

.....

.....

[3]

4 This is a section of program code.

```
1 Total = 100.00
2 PRINT 'Enter the height of each member of your class, one at a
  time, when prompted'
3 FOR Count = 1 TO 30
4   PRINT 'Enter a height in metres'
5   INPUT Height
6   Total = Total + Height
7   PRINT Total / 30
8   Count = Count + 1
9 NEXT Count
```

(a) There are **three** errors in this code.

State the line numbers that contain the errors and describe how to correct each error.

Error 1 .....

.....

.....

Error 2 .....

.....

.....

Error 3 .....

.....

.....

[3]

(b) State the purpose of this program.

.....

.....

.....[1]

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5 The algorithm allows a number to be entered. It then calculates and outputs the next number in the mathematical series.

```
Fib ← 1
Prev2 ← 0
Prev1 ← 1
INPUT Number
IF Number = 0
    THEN Fib ← 0
ENDIF
WHILE Number > 2
    Fib ← Prev2 + Prev1
    Prev2 ← Prev1
    Prev1 ← Fib
    Number ← Number - 1
ENDWHILE
OUTPUT Fib
```

(a) Complete the trace table for the input data: **7**

Fib	Prev2	Prev1	Number	OUTPUT

[4]

(b) Complete the trace table for the input data: **2**

Fib	Prev2	Prev1	Number	OUTPUT

[2]

6 An online fruit tree specialist sells fruit trees in various sizes. A database table, TREETAB, shows the tree type and, for each size, the price and whether they are in stock.

Tree Type	Size1	Size1 In	Size2	Size2 In	Size3	Size3 In
Apple	10.95	Yes	14.95	Yes	29.95	Yes
Apple	12.95	Yes	14.95	Yes	29.95	Yes
Cherry	24.95	No	34.95	No	59.95	Yes
Fig	19.95	Yes	29.95	No	49.95	Yes
Guava	19.95	No	29.95	No	59.95	No
Nectarine	8.50	Yes	11.95	Yes	19.95	Yes
Olive	19.95	No	39.95	Yes	59.95	Yes
Peach	9.25	No	11.95	Yes	19.95	Yes
Pear	10.95	Yes	14.95	Yes	29.95	Yes
Plum	8.95	Yes	11.95	Yes	19.95	Yes
Pomegranate	12.95	No	18.95	Yes	34.95	No
Quince	34.95	Yes	44.95	Yes	84.95	No

(a) State whether any of the fields shown would be suitable as a primary key.

.....

Explain your answer .....

.....

.....

[2]

(b) Complete the table to show the most appropriate data type for each of the fields based on the data shown in the table at the start of question 6.

Field	Data type
Tree Type	
Size3	
Size2 In	

[3]

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(c) Show the output that would be given by this query-by-example.

Field:	Tree Type	Size1	Size1 In		
Table:	TREETAB	TREETAB	TREETAB		
Sort:		Descending			
Show:	☑	☑	☑	☐	☐
Criteria:		<10.00			
or:					

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.....[4]

(d) Using the following query-by-example grid, write a query to identify all types of the fruit trees that are out of stock for all three sizes. Make sure the type of the tree and the various 'in stock' fields are shown. The trees should be listed in alphabetical order by type.

Field:					
Table:					
Sort:					
Show:	☐	☐	☐	☐	☐
Criteria:					
or:					

[4]

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